12 **(a)** State what is meant by luminosity of a star.

.....

..... [1]

(b) The luminosity of the Sun is 3.83×10^{26} W. The distance between the Earth and the Sun is 1.51×10^{11} m.

Calculate the radiant flux intensity F of the Sun at the Earth. Give a unit with your answer.

F = ... unit ..... [2]

(c) Use data from **(b)** to calculate the mass that is converted into energy every second in the Sun.

mass = ... kg [1]

(d) The radius of the Sun is 6.96×10^{8} m.

Show that the temperature T of the surface of the Sun is 5770 K.

[1]

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(e) The wavelength λ_{max} of light for which the maximum rate of emission occurs from the Sun is 5.00×10^{-7} m.
The temperature of the surface of the star Sirius is 9940 K.

Use information from **(d)** to determine the wavelength of light for which the maximum rate of emission occurs from Sirius.

wavelength = ... m [2]

[Total: 7]